

DATA FLOW DIAGRAMS (DFD) SPECIFICATION

SMARTSHOP

Level 0 (Context Diagram) and Level 1 (Functional Decomposition)

1. DFD NOTATION STANDARD (Yourdon & DeMarco / Gane & Sarson)

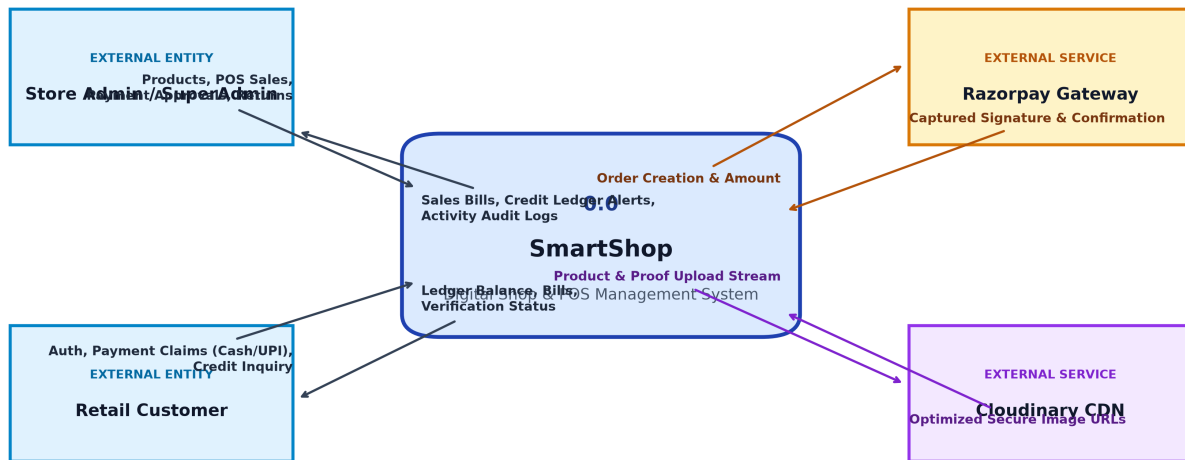
A Data Flow Diagram (DFD) models the graphical transformation of data through an information system. The SmartShop DFD adheres to academic engineering standards utilizing four primary symbols:

Symbol	Notation Element	Representation in SmartShop
Rectangle	External Entity	Actors outside system boundary: Store Admin, Retail Customer, Razorpay, Cloudinary
Rounded Box / Circle	Process / Function	Transformational operations: 0.0 (System), 1.0 (Auth), 3.0 (POS Sales), 4.0 (Khata), etc.
Open Parallel Lines	Data Store	Persistent MongoDB collections: D1 (Users) through D7 (Returns)
Directed Arrow	Data Flow	Flow of inputs, outputs, credentials, invoice data, and receipts

2. DFD LEVEL 0 — CONTEXT LEVEL DIAGRAM

The Level 0 context diagram defines the boundary of the SmartShop system, showing external actors and major data flows interacting with the single top-level process **0.0 SmartShop System**.

SmartShop — DFD Level 0 (Context Diagram)



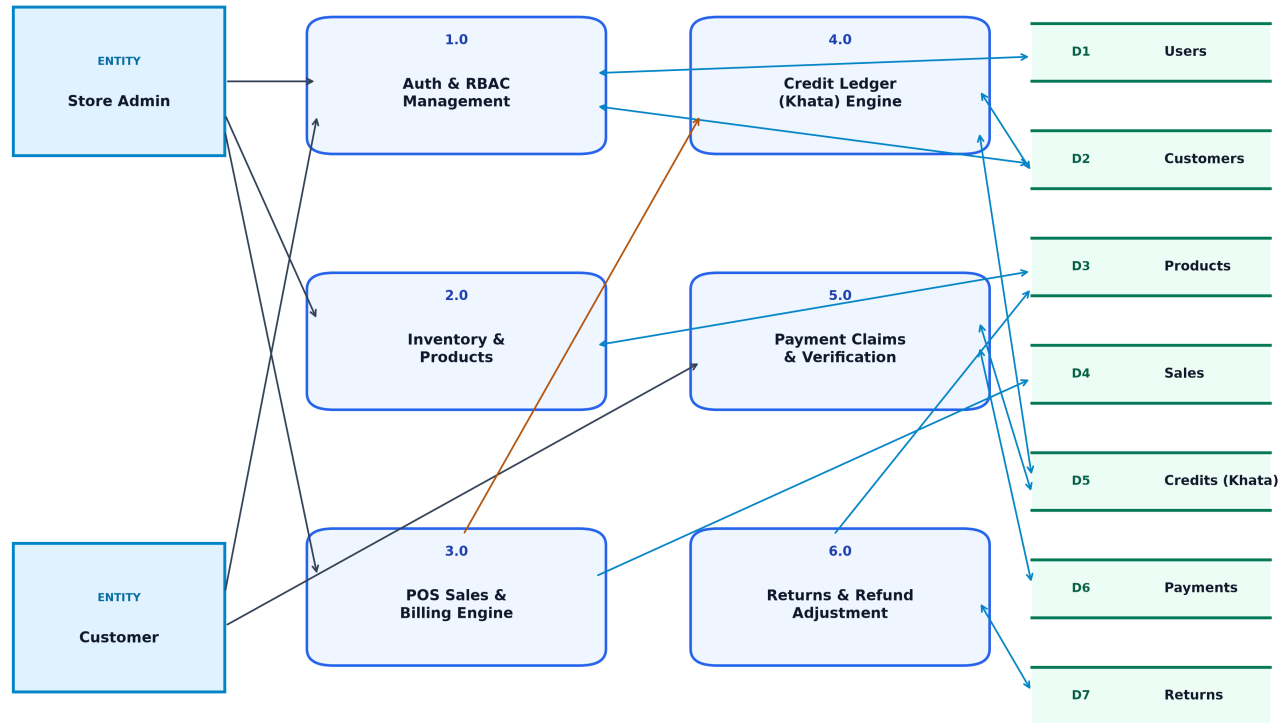
Detailed Data Inflows & Outflows:

- **Store Admin -> System:** Product catalog details, POS cart items, payment claim approval/rejections, return orders, customer borrow limits.
- **System -> Store Admin:** Real-time sales invoices, pending credit balance reports, activity audit trail, low-stock alerts.
- **Customer -> System:** Registration/login credentials, cash payment claims, UPI transaction IDs and screenshot proofs, online payment tokens.
- **System -> Customer:** Itemized purchase receipts, live Khata debt balance, payment verification receipts.
- **System <-> Razorpay / Cloudinary:** Online checkout order IDs & captured signature verification; Cloudinary CDN media upload streams & secure image asset URLs.

3. DFD LEVEL 1 — FUNCTIONAL DECOMPOSITION DIAGRAM

The Level 1 DFD decomposes the central system process into **6 core functional sub-processes** and maps their read/write interactions with **7 persistent database collections (D1–D7)**.

SmartShop — DFD Level 1 (Functional Decomposition Diagram)



Process Decomposition & Data Store Mapping:

- **1.0 User Authentication & RBAC:** Authenticates customers and admins. Reads/writes **D1 (Users)** and **D2 (Customers)**.
- **2.0 Product & Inventory Management:** Manages product CRUD, prices, units, and Cloudinary image assets in **D3 (Products)**.
- **3.0 POS Sales & Billing Engine:** Processes counter bills. Deducts stock in **D3**, creates invoices in **D4 (Sales)**, and triggers credit creation in **4.0**.
- **4.0 Credit / Digital Khata Engine:** Calculates trust scores, evaluates borrow limits, and updates debt records in **D2 (Customers)** and **D5 (Credits)**.
- **5.0 Payment Reconciliation:** Records cash/UPI/Razorpay claims in **D6 (Payments)** and decrements customer pending dues atomically in **D5** and **D2**.
- **6.0 Returns & Refund Engine:** Restores product quantities in **D3 (Products)**, records return slips in **D7 (Returns)**, and offsets outstanding debt in **D5 (Credits)**.